1 Worked example: estimability in a two-receiver network

This section works the rank deficiency and minimum-constraint datum of the minimum-constraint datum section of the Network-mode document through numerically on the smallest network that shows the structure: two receivers $\mathcal{R} = \{A, B\}$ and four satellites $i \in \{1, 2, 3, 4\}$. Everything not needed for the estimability argument is removed. Positions are known, so the geometric range drops out; a single frequency is used; the troposphere and ionosphere are absent; the data are noise-free. Every number below is therefore exact, and the gauge freedoms are visible by inspection. The subject of the example is the null space of the system, not its size.

1.1 Code only: clocks and one gauge freedom

With geometry removed, the single-frequency pseudorange of the observation-equations section collapses to a difference of two clock lumps,

$$P_r^i \;=\; a_r \;-\; g^i, \qquad a_r := c\,\mathrm{d}t_r + B_r, \quad g^i := c\,\mathrm{d}t^i + B^i, \tag{1}$$

where a_r is the receiver clock and g^i the satellite clock. On a single frequency the receiver and satellite *code biases* B_r, B^i are inseparable from the respective clocks, so each "clock" here is a clock + code-bias lump. With one band, the IF-vs-DCB split of the satellite phase-bias section of the Network-mode document does not arise. The system has $|\mathcal{R}| + |\{i\}| = 2 + 4 = 6$ unknowns and $2 \times 4 = 8$ observations.

The ground truth (metres) is

$$a_A = 5, \quad a_B = 3, \qquad g^1 = 2, \quad g^2 = -1, \quad g^3 = 4, \quad g^4 = 0.5,$$

and Equation (1) maps it to the eight pseudoranges

P_r^i	$i=1$	$i=2$	$i=3$	$i=4$
$r=A$	3	6	1	4.5
$r=B$	1	4	−1	2.5

The deficiency from two directions. *(i) Invariance.* Adding a constant κ to all six unknowns, $a_r \mapsto a_r + \kappa$ and $g^i \mapsto g^i + \kappa$, leaves every difference $a_r - g^i$, and therefore every observation, unchanged. The truth above and the shifted set $\{15, 13, 12, 9, 14, 10.5\}$ fit the data identically. The data determine clock differences only, never absolute clocks.

(ii) Singular normals. With the unknowns ordered as $\boldsymbol{x} = [\,a_A, a_B, g^1, g^2, g^3, g^4\,]^{\mathrm{T}}$ and unit weights, the normal matrix $\boldsymbol{N} = \boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}$ is

$$\boldsymbol{N} = \begin{bmatrix} 4 & 0 & -1 & -1 & -1 & -1 \\ 0 & 4 & -1 & -1 & -1 & -1 \\ -1 & -1 & 2 & 0 & 0 & 0 \\ -1 & -1 & 0 & 2 & 0 & 0 \\ -1 & -1 & 0 & 0 & 2 & 0 \\ -1 & -1 & 0 & 0 & 0 & 2 \end{bmatrix}, \qquad \boldsymbol{N}\,\boldsymbol{1} = \boldsymbol{0}, \tag{2}$$

a 6×6 matrix of rank 5. The all-ones vector $\boldsymbol{1} = [1, 1, 1, 1, 1, 1]^{\mathrm{T}}$ is an exact null vector (every row sums to zero), and it is the common-shift direction of view (i). The system $\boldsymbol{N}\boldsymbol{x} = \boldsymbol{u}$ has no unique solution without one added constraint, a *datum*.

Two datum choices, same fit. Pinning one clock removes the gauge. Two natural choices give different parameter values, and both fit the eight observations exactly:

datum (S-basis)	a_A	a_B	g^1	g^2	g^3	g^4
$a_A \equiv 0$ (reference receiver)	0	−2	−3	−6	−1	−4.5
$g^1 \equiv 0$ (reference satellite)	3	1	0	−3	2	−1.5

Under $a_A \equiv 0$, every satellite-clock estimate comes out as $\hat{g}^i = g^i - a_A^{\text{true}}$: the reference receiver's clock is absorbed into all of them. The estimated "satellite clock" is the physical clock shifted by the chosen datum. This offset is the *lumping*, and the user must account for it. For this reason a producer declares its datum (the closed-loop section of the Network-mode document).

What is invariant. The estimable quantities (combinations annihilated by the null vector $\mathbf{1}$, i.e. differences) are identical in both bases:

estimable quantity	$a_A \equiv 0$	$g^1 \equiv 0$	(truth)
$a_A - a_B$	2	2	2
$g^2 - g^1$	−3	−3	−3
$g^3 - g^1$	2	2	2
$a_B - g^4$	2.5	2.5	$(= P_B^4)$

The absolute values are a convention; the differences are physical. This is the S-system of [1] in six parameters.

Remark 1 (Why the producer cannot difference). The double difference of this code-only model is $\nabla\Delta P_{AB}^{i,1} = (P_A^1 - P_A^i) - (P_B^1 - P_B^i)$. The between-satellite step gives $g^i - g^1$ for both receivers, so the between-receiver step yields exactly $\mathbf{0}$. Double differencing eliminates the entire model: DD removes clocks, and here the model contains nothing else (the differencing section). A DD user estimates nothing. The network producer instead keeps the clocks and pins one datum. This degeneracy is, in miniature, the reason a network producer exists.

1.2 Adding carrier phase: the satellite phase bias needs the network

The single-frequency carrier phase, with the same clock lumps, is

$$\Phi_r^i \;=\; a_r - g^i \;+\; b^i \;+\; N_r^i, \tag{3}$$

where b^i is the satellite phase bias, the producer's target, and N_r^i is the float ambiguity (in metres). The receiver phase bias and the receiver code-minus-phase offset are constant per receiver and are absorbed into N_r^i. The satellite bias b^i stays explicit because the producer emits it. The truth is extended with

$$b^1 = 0.10, \qquad b^2 = -0.20, \qquad b^3 = 0.05, \qquad b^4 = 0.15,$$
$$N_A^i : 10,\ -5,\ 20,\ 8, \qquad N_B^i : 12,\ -3,\ 18,\ 9,$$

giving the eight carrier-phase observations

Φ_r^i	$i=1$	$i=2$	$i=3$	$i=4$
$r = A$	13.10	0.80	21.05	12.65
$r = B$	13.10	0.80	17.05	11.65

Counting: one receiver vs. two. The code fixes the clocks, so per satellite the phase contributes $b^i + N_r^i$. With one receiver this is a single equation $b^i + N_A^i$ in two unknowns. The satellite phase bias is then inseparable from that receiver's ambiguity, and the equation carries no redundancy. With two receivers there are two equations,

$$b^i + N_A^i, \qquad b^i + N_B^i, \tag{4}$$

in three unknowns (b^i, N_A^i, N_B^i): rank 2, one deficiency per satellite. The estimable two-dimensional space contains the *between-receiver single-difference ambiguity* $\nabla N_{AB}^i = N_A^i - N_B^i$. This quantity does not exist with one receiver, and it is the integer a user resolves (the differencing section). The second receiver gives b^i an identity shared across stations.

Datum and solution. The joint code + phase system has a 1-dimensional clock gauge (as before) plus one phase-bias↔ambiguity gauge per satellite: $b^i \mapsto b^i + c_i$, $N_r^i \mapsto N_r^i - c_i$ for all r. The minimum-constraint datum used here has two parts. The first is the clock pin $a_A \equiv 0$. It is S-transform-equivalent to the reference-*satellite* clock pin $g^1 \equiv 0$ that the estimator's S-basis declares (the minimum-constraint datum section of the Network-mode document): in its default pin mode the shared-block gauge pins one satellite clock per constellation, not a receiver clock (the optional `network.clock_datum_sum_to_zero` and `network.clock_datum_broadcast_anchored` modes replace the pin). The second part is the set of reference-receiver ambiguity pins $N_A^i \equiv 0$ for every i: the local datum on the pivot receiver, folded into station A's rows. The solution is

	$i=1$	$i=2$	$i=3$	$i=4$
emitted phase bias $\bar{b}^i = b^i + N_A^i$	10.10	−5.20	20.05	8.15
$\hat{N}_A^i$ (datum)	0	0	0	0
$\hat{N}_B^i = N_B^i - N_A^i$	2	2	−2	1

with the clocks unchanged from Section 1.1. The fit checks directly, e.g. $\hat{\Phi}_B^3 = a_B - g^3 + \bar{b}^3 + \hat{N}_B^3 = -2 - (-1) + 20.05 + (-2) = 17.05$, matching the observation.

Reading the result. The solution shows two properties.

- *The lump dominates the emitted bias.* The physical satellite phase bias of satellite 1 is $b^1 = 0.10$ m, while the emitted estimable bias is $\bar{b}^1 = 10.10$ m: it has absorbed receiver A's ambiguity (10 m) for that satellite. This is Odijk's $\bar{b}^i = b^i +$ (pivot-receiver ambiguity) (the satellite phase-bias section of the Network-mode document, Eq. 4/9) in numbers. A consumer recovers integers only by removing the same lumped quantity, so the producer datum and the consumer integer recovery must belong to one S-system.
- *The network makes the integer estimable.* The estimable $\hat{N}_B^i = N_B^i - N_A^i = \nabla N_{AB}^i$ is the between-receiver single-difference ambiguity, the quantity an RTK user fixes to an integer. It is absent from a single-receiver problem; the second station introduces it. The satellite phase bias becomes a definite, transferable number at the cost of pinning one reference ambiguity, and it carries that pin.

Up to the geometry and the second frequency it omits, this four-satellite, two-receiver instance has the structure of the synthetic network on which the estimator core is tested. The `RecoversSyntheticTruthUpToSTransform` test (`tests/test_network_estimator.cpp`) builds rows of this shape (four receivers and five satellites, from a seeded random truth), pins the

reference-satellite clock datum $g^1 \equiv 0$ (S-transform-equivalent to the $a_A \equiv 0$ used above) through `NetworkEstimator::solve`, and pins the pivot receiver's ambiguities with tight-prior rows in that station's block. It asserts recovery of the satellite-side parameters up to the declared S-transform: the lumped $\bar{b}^i$ and the single-difference ambiguities above, not the unobservable physical b^i.

References

[1] P. J. G. Teunissen. "Generalized inverses, adjustment, the datum problem and S-transformations". In: *Optimization and Design of Geodetic Networks* (1985), pp. 11–55.